List of Corrections

Would some examples of each existing SIP be useful? 31
Anything else about perceptual scalars? Maybe depth? 33
Very short. Is there anything more interesting to say? 33

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Abstract

Spatial information is crucial to the listener's experience of music, and stereophonic recordings have long been the standard for capturing this spatiality. Recent advancements in computing, coupled with the availability of new, very large musical corpora, have created opportunities for large-scale quantitative analysis of the stereo image of recorded music. This licentiate thesis addresses a key gap in current research: direct analysis of raw music recordings to describe the stereo image. It will explore existing methods, assess their effectiveness, and demonstrate potential use cases for spatial analysis within audio reproduction, transmission and musical performance.

Dedicated to the one guy at Uppsala University who made a latex template 14 year ago

List of papers

This thesis is based on the following papers, which are referred to in the text by their roman numerals.

I A. Franklin et al., 'Development of a Web-Based Real-Time Distributed Reverberation Chamber for Live Concert Applications,' *Journal of The Audio Engineering Society*, vol. 73, no. 12, pp. 872–881, DOI: 10.17743/jaes.2022.0245

II R. Mitic and A. Rossholm, 'Towards an objective comparison of panning feature algorithms for unsupervised learning,' in *Proceedings of the 28th International Conference on Digital Audio Effects (DAFx25)*, Ancona, IT. [Online]. Available: https://dafx.de/paper-archive/2025/DAFx25_paper_26.pdf

III R. Mitic et al., 'Perceptual Evaluation of Binaurally Panned Speech and Music Mixtures,' *Acta Acustica*, in review

IV R. Mitic and A. Rossholm, 'A Systematic Literature Review of Methods to Describe the Stereo Image of Recorded Music,' *EURASIP Journal on Audio, Speech, and Music Processing*, accepted. DOI: 10.1186/s13636-026-00463-4

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1. Introduction

It is a widely accepted truth in the world of music production and recording that spatial information in audio is crucial to listeners' sense of immersion and connection with the music. To capture that spatiality, stereophonic recording and reproduction has been the standard format since it gained popularity in the 1950s [5]. The spatial analysis of subsequent decades worth of stereo recordings presents a unique opportunity to study human musicality, as each recording reflects and encodes the creative choices made by the musicians and recordists involved.

Within the music community, it is common to speak about spatiality in a qualitative sense, using terms such as 'stereo image' and 'stereo field' to encompass all audible spatial aspects. There have been several attempts to translate these discussions in a quantitive domain [6–9], all of which were performed in an era when musical datasets were hand-curated/-labelled, fairly small (1,000 - 2,000 tracks), and – most notably – private. However, in recent years new musical corpora have become available, e.g. MTG-Jamendo [10] and Free Music Archive (FMA) [11], which are both orders of magnitude larger than their predecessors, containing approximately 55,000 and 100,000 tracks respectively. In addition, computing techniques in Machine Learning (ML) and Deep Learning (DL) have progressed to the point where it is now possible to learn extremely high-level features on these vast datasets, thereby making sense of the human creativity represented in the recordings.

A typical use case for spatial descriptors that has emerged both in academia and industry is classification, and in particular, auto-tagging of musical genre/performer [12, 13]. Such features can be used in music similarity and recommendation tasks [14, 15]. It has also been shown that stereo field classification of subsections within a larger work can act as a precursor to other tasks such as segmentation, audio enhancement, upmixing, and source separation [16–18]. The latter is very relevant to the audio distribution industry, which is currently experimenting with newer audio formats such as ambisonic and object-based audio. While these new formats offer exciting opportunities for both distributors and listeners, there remains a huge back-catalogue of stereophonic recordings that could benefit from a high-level understanding of their spatial content.

Historically, binaural audio (which is a subclass of stereo) has been of particular interest because its relevance to human hearing. Thus, it is extremely widely studied, and the underlying fundamental techniques are well-established [19]. Yet there remains a gap in current research regarding direct

analysis of raw music recordings, and in particular, describing the stereo image of said recordings at a high level. This thesis aims to fill that gap. It will firstly demonstrate use cases where the spatial analysis of stereo audio has the opportunity to solve real-world problems in audio transmission, dissemination, and reproduction. Accordingly, it will gather existing methods to describe stereo image and assess their abilities to detect the spatial aspects of a variety of musical audio sources.

2. Research Questions and Contributions

The underlying purpose of the research documented in this thesis relates to Music Information Retrieval (MIR), and in particular to generating an understanding of the links between human musical creativity and musical output. While it might be possible for the creative people involved to explain how they make their music, it is far more objective to analyse their output as a representation of their musical ideas.

2.1 Research Questions

Given the aforementioned context, the overarching research question (RQ) can be formed as:

How do music producers use stereo field?

...which can be broken into several smaller problems:

RQ_1: *How does the stereo image of a recording relate to real-world musical knowledge?*
RQ_2: *What methods can be used to describe the stereo image of music recording?*

One could argue that the overarching question might be answered by interviewing music producers, but this would quickly run into a problem of scale. The two smaller RQs were therefore designed to allow the overarching question to be answered by objective analysis on musical recordings (which are abundant nowadays).

2.2 Research contributions

This section presents a summary of the research contributions achieved by the 4 included papers, and explains their relationship to the RQs.

2.2.1 An understanding of the musical uses of stereo field

The design and execution of the Auxtrument led to a direct understanding of how a musical can use stereo field to convey a message that would otherwise

be difficult to communicate. With respect to the overarching research question, the act of creating musical output gives a small window into how a music producer might use stereo field. Paper I was a retrospective account of the process of building the Auxtrument, which additionally provides a technical contribution to any musicians wishing to make a similar performance in the future. The paper also provided suggestions for future iterations of Auxtrument-style instruments and the standardized Application Programming Interfaces (APIs) which are built into web browsers, providing a basis for future work.

The Orkester framework and software library presented the opportunity to apply a more technical approach to RQ_1. The exploration of stereo field in existing recordings furthered the knowledge gained from an individual musical project by allowing us to analytically compare stereo fields created by many producers. This allowed us to establish an understanding of the range and variation that stereo field can embody.

2.2.2 Signal processing techniques relating to stereo field

The work on Orkester and the resulting paper II contribute an understanding of how some established techniques for measuring stereo field react to existing recorded music. Moreover, the understanding is not limited to the techniques specified in paper II since the development of Orkester involved a small-scale literature search and subsequent implementation of many other techniques, not just those that were eventually included in the paper.

In order to fully answer RQ_2, a much more extensive Systematic Literature Review (SLR) (paper IV) was performed with two intentions. First and foremost, the intention was to conclusively find all existing mathematical techniques that describe stereo field. In addition, there was a goal to find 'cross-pollination' opportunities from disparate areas of audio research which nonetheless derive spatial cues from audio signals. In doing so, it was possible to form a 'general model' of Stereo Image Profiles (SIPs), thereby making it easier to for future researchers to find or create a SIP that suits their specific use case.

2.2.3 Psychoacoustics of mixed music and speech

The perceptual listening test described in paper III provided a new perspective on a classic problem in psychoacoustics research. While there is extensive literature describing how people perceive music or speech that has been intentionally masked by another sound, our experiment used music and speech that were intended to complement each other. The results firstly suggested that while people prefer speech that is panned towards the centre, there was an opportunity to increase the audible pleasantness by using very subtle panning, and moreover that said panning might be automated by analysis of the stereo

field (thereby helping to answer RQ_1). Furthermore, it became clear that the intelligibility of speech was paramount to pleasantness, despite the original assumption that the music and speech complemented each other.

2.2.4 Setup for future work on large datasets of audio

In regard to the future work described in chapter 7, the SLR provided both a list of relevant audio datasets, and a many standardized implementations of mathematical techniques that extract spatial information from audio. Paper II also presented a good opportunity to assess the computational complexity of the algorithms implemented in Orkester, thereby highlighting some opportunities for optimization when they are applied to extremely large corpora.

3. Background

This chapter provides the fundamental background knowledge that underpins the work included in the thesis. It covers human hearing, music creation processes, and digital audio signal theory.

3.1 The mechanics of human hearing

Lindsay's book *Human Information Processing; an Introduction to Psychology* [20] gives a detailed description of how the human ear is constructed and how it enables us to hear. The relevant parts are summarized below.

The ear contains three major sections, loosely labelled the 'outer', 'middle', and 'inner' ear. When air vibrations (i.e. sound waves) flow into the outer ear, mechanical coupling in the middle ear transmits vibrations to the cochlea – the major organ of the inner ear. The cochlea is a conical tube which is coiled up into a spiral, filled with an incompressible fluid, and covered on either end by a membrane. Through the centre of the tube runs the basilar membrane, which is a covered in hairs named cilia. When vibrations are mechanically transmitted to the base of the cochlea, the internal fluid responds by transmitting those vibrations to the basilar membrane, and the cilia will bend and twist accordingly. The cilia are linked to nerve fibres, and will initiate electrical pulses when manipulated. These electrical pulses are sent to the brain and interpreted as sounds.

Along the length of the cochlea, the cilia have different 'critical frequencies' that they respond to. The critical frequency increases logarithmically as one moves from the base (the widest section) to the apex (the narrowest section) of the cochlea. The cochlea therefore acts to convert time-domain sound waves into time- and frequency-dependent electrical excitations, thereby allowing humans to perceive different frequencies simultaneously. Figure 3.1 shows a tonotopic diagram of the cochlea.

3.2 Directional hearing

Following from section 3.1, this section will discuss humans' ability to discern the direction or location of a sound source.

The psychoacoustic theory behind spatial hearing has been an active area of research since the mid 20th century, with early examples including the Jeffress

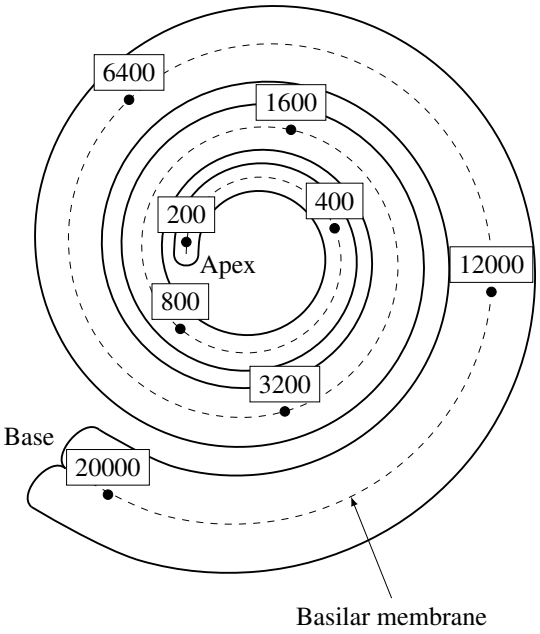


Figure 3.1. Tonotopic diagram of the human cochlea, showing exemplary positions of critical frequencies in Hz.

model in which time differences between sound events at each ear contribute to a person's ability to locate sounds in space [21]. The work of Lindemann a few decades later expanded on this idea by translating the earlier theory to a Digital Signal Processing (DSP) context, and establishing the Interaural Level Difference (ILD), Interaural Time Difference (ITD), and Interaural Cross Correlation (IACC) as important measures to describe the spatial aspects of binaural signals in a quantifiable way [22].

Put simply, humans are able to perceive the incoming direction of a sound by observing both the amplitude difference and time difference of coincident sound waves arriving at the left and right ears. Figure fig. 3.2 shows how these two properties enable directional hearing. Firstly, the head itself creates a physical barrier that will attenuate sound waves at one ear more-so that at the other ear. Secondly, the difference in distance between the sound source and each ear will create a time delay between the received signals. Over time a person will learn to associate patterns in ITD/ILD with the physical location of the sound source.

The relative importance of ILD and ITD differ depending on the frequency of the incoming sound. The reasons for this are twofold. Firstly, a person's inter-aural distance (the physical distance between their ears) must be longer than the wavelength of the incoming sound wave in order for them to detect the phase difference between wave peaks. In this case, it is often refered to

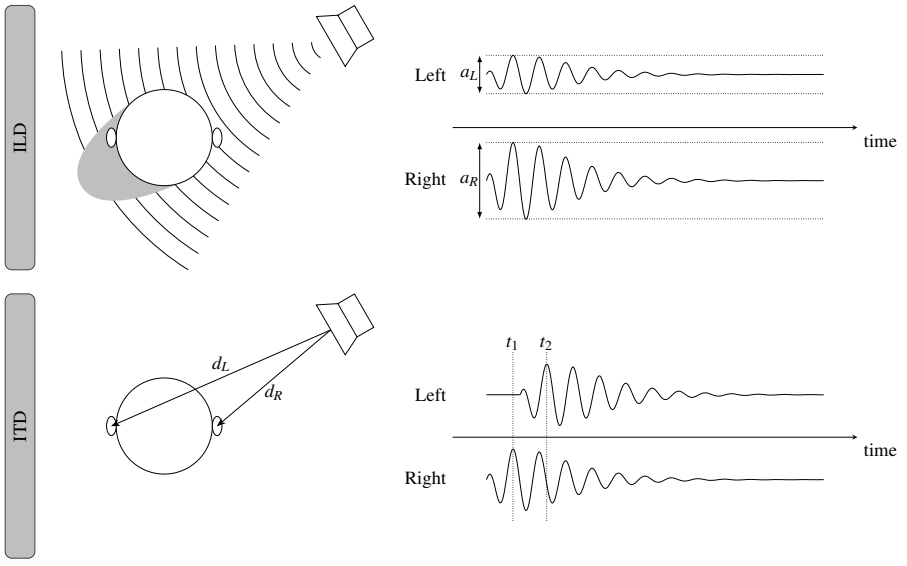


Figure 3.2. Illustration of how humans perceive the direction of an incoming sound. The physical barrier of the head causes an Interaural Level Difference (ILD) between sound waves received at the two ears ($a_R - a_L$). The difference in distance from the source to the ears ($d_R - d_L$) causes an Interaural Time Difference (ITD) between the two ear ($t_2 - t_1$).

as Interaural Phase Difference (IPD), which accounts for the centre frequency of the critical band. Secondly the cilia that line the basilar membrane inside the cochlea can only detect the fine structure of a sound wave up until roughly 1.6kHz [22]. Lindemann gives a rule of thumb that time differences of less than 7.5ms are heard as a single auditory event (for which direction could be inferred), and delays above 21ms are heard as uncorrelated events. Hence, at 1.6kHz and below ITD is more important, whereas above 1.6kHz ILD is more important. Furthermore, Blauert's subsequent work found firstly that both ILD and ITD are relevant within the frequency band surrounding 1.6kHz, and secondly that the angle of incidence of the sound wave plays a role in how the cues are perceived [23]. Despite this, Lindemann's rules of thumb are still broadly accepted, even being encoded into international standards such as [24]. As an example, fig. 3.3 shows two recordings (a low-frequency kick drum and high-frequency hi-hat) that originate from the right-hand side of a listener. The high-frequency hi-hat has a large ILD and ambiguous IPD, whereas the low-frequency kick drum has negligible ILD, but an interpretable IPD.

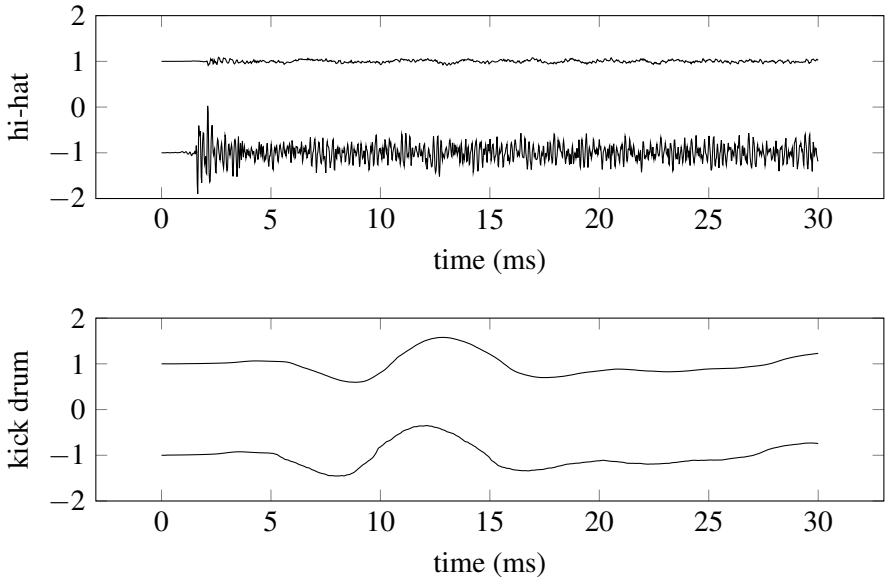


Figure 3.3. Example sounds coming from the right-hand side, recorded binaurally. High-frequency content (above) has a large ILD and ambiguous IPD, whereas low-frequency content (below) has negligible ILD, but an interpretable IPD.

3.3 Loudness perception

Humans perceive the loudness of sound, not in terms of the absolute amplitude of the sound wave, but the *power* of the sound wave. Mathematically speaking, perceived loudness is proportional to the Root Mean Squared (RMS) Sound Pressure Level (SPL) raised to an exponent [25]. Observing that the calculation of RMS naturally involves taking the square of the original time function (see eqs. (3.1) and (3.2) for the continuous and discrete forms respectively), it follows multiplication of a signal by $\sqrt{2}$ will result in double perceived loudness. Conversely, multiplying a signal by $\frac{1}{\sqrt{2}}$ will halve the perceived loudness.

$$f_{RMS}(t) = \sqrt{\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} [f(t)^2] dt} \tag{3.1}$$

$$f_{RMS}[n] = \sqrt{\frac{1}{N} \sum_{n=0}^{N} f[n]} \tag{3.2}$$

3.4 Panning

The concepts of directional hearing are applied in music production to make sound sources appear as though they originate from a specific direction or position, often referred to as the 'phantom position'. In recording studio vernacular, this is call 'panning'. In accordance with the two signal differences a that humans are able to detect, there are also two basic types of panning – amplitude panning and time panning. By way of example, fig. 3.4 shows how the phantom position can be moved anywhere in between the speakers that are reproducing the sound by applying different amplitude gains to the left and right channel of a stereo recording.

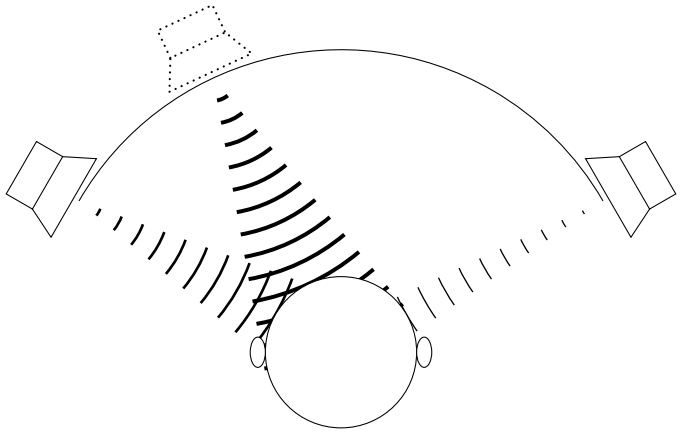


Figure 3.4. Perception of a phantom sound source created by amplitude panning. The phantom position is represented by a dotted loudspeaker. The sound wave from the left speaker has higher amplitude than that of the right speaker, thereby creating a phantom source which appears to come from the left.

Time panning (otherwise know as the Haas effect) exploits the phenomenon that the first ear to hear a sound generally dictates the direction that it appeared to come from. The process is to simply delay the signal in one channel by a time value within the 'Haas window' (typically less than 40ms). The undelayed channel will then dictate the phantom direction of the original source. In music production is used both for panning and to add spaciousness to a recording [26, 27].

Another frequently used technique is 'binaural panning', which assigns combinations of amplitude and time difference to each stereo channel. This is typically achieved by applying a Head-Related Impulse Response (HRIR) to the monophonic audio signal. An HRIR is a pair of impulse responses recorded with microphones placed in the ears of either a real person or a dummy head. To obtain an HRIR, a known excitation signal is played from a specific position in a room, and is recorded with said binaural microphones (see fig. 3.5). The HRIR can then be convolved with any other monophonic record-

ing to make it appear as though the recording came from the position of the excitation source.

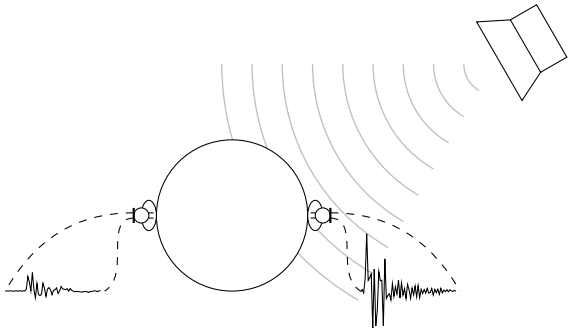


Figure 3.5. Illustration of a binaural HRIR being recorded.

In traditional recording studio setups, the recording engineer or producer will use one or more microphones to capture an original sound source. These singular sound sources are commonly called 'stems'. During the production process, the stems may be panned to create a sense of space in the final mix. In the case that the stem was recorded using two microphones, the stem would be called a 'stereo stem', and the term 'panning' is usually replaced by 'balance', and only amplitude panning (as opposed to time or binaural panning) would be applied.

Amplitude panning is the most common panning technique is traditional music production [28]. It is associated with so-called 'panning laws', which govern how the amplitude is changed in each channel according the desired pan position [27]. One example is the 'energy-preserving pan law' (sometimes called the '-3dB pan law') in which the squared sum of the channel gains always equals 1. The gains of the left and right channel are calculated by $g_L = \cos(\frac{\pi}{2}a)$ and $g_R = \sin(\frac{\pi}{2}a)$ respectively where a is the panning position in the range $[0,1]$. Figure 3.6 shows how the channel gains relate to the panning position. The total perceived loudness will be equal at all panning positions.

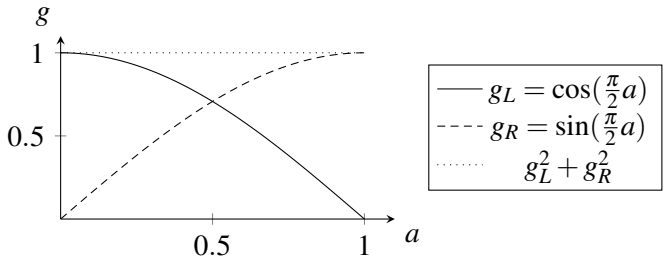


Figure 3.6. Equal power panning. g is the gain applied to the channel and a is the panning factor ranging from 0 (fully left) to 1 (fully right).

3.5 Stereo image

The term ‘stereo image’ is used to describe the overall distribution of sound across the horizontal plane after panning and any other production techniques have been applied. One may also use terms like ‘stereo field’, ‘soundstage’, or ‘panorama’, which all refer to the same thing. They are usually used in qualitative descriptions such as ‘big’, ‘wide’, and ‘ambient’ rather than quantitative measures [29]. When mixing a musical recording, a producer will pan individual stems to their desired position within the stereo image. Those stems which are recorded in stereo in fact have their own stereo image, which can also be controlled by the producer. The combination of all stems into a final stereo mix, constitutes the overall stereo image.

Figure 3.7 shows a simple example stereo image, as perceived by the producer. It contains two mono stems – vocals and violin, and one stereo stem – piano. There are some common tropes seen throughout many productions, some of which are shown in the diagram. Firstly, vocals are recorded in mono and panned to the centre. Secondly, since the piano is panned slightly left, the violin is panned roughly the same amount to the right in order to achieve a balanced mix. Thirdly, all stems are placed in the stereo image in a way that emulates a natural performance on a stage, and to avoid separate instruments masking each other. Note that while these practices are typical, they are purely an artistic choice on behalf of the producer [29]; Recordings breaking the trend are not difficult to find.

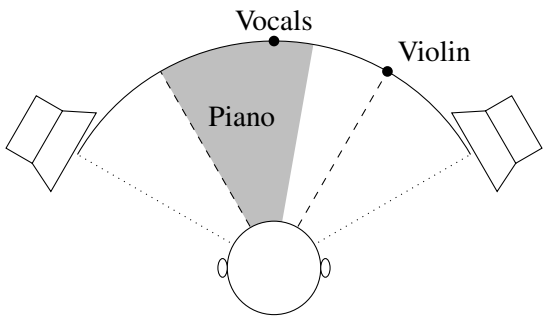


Figure 3.7. Example stereo image created by a music producer. Mono stems are shown as black dots and stereo stems as grey fills. Dashed lines represent the extent of the stereo image. Dotted lines represent the maximum boundaries of the possible stereo image.

Describing the stereo image of a recording has its roots in the acoustics domain, in which typical use cases are the analysis of concert halls and music reproduction systems. In these cases, the previously mentioned low-level spatial cues typically form the basis of the analysis. Berg and Rumsey [30] give a long list of typical perceptual measures that can be derived from the low-level cues, although later literature has focused on the measures of Apparent Source

Width (ASW) and Listener Envelopment (LEV). However, as stated by Sarroff and Bello [31], the task of describing the spatial aspects of recorded music is inherently different to that of acoustical signals since said recordings may exhibit spatial aspects that cannot occur in physical acoustic space.

3.6 Music production

A typical setup for a music producer is to use a Digital Audio Workstation (DAW) - graphical software that allows them to record and edit audio. There are many different DAWs available, but one such example (Reaper) is shown in fig. 3.8a. In particular, it shows a single recorded audio track which has it's panning automated. It also contains a single effects track which applies a stereo reverb effect to the recorded audio. This effects track has its 'dry/wet' mix parameter automated, which controls the ratio of non-processed to processed sound. In a more complete recording session, the DAW would contain many more audio and effects tracks, each with their own panning control.

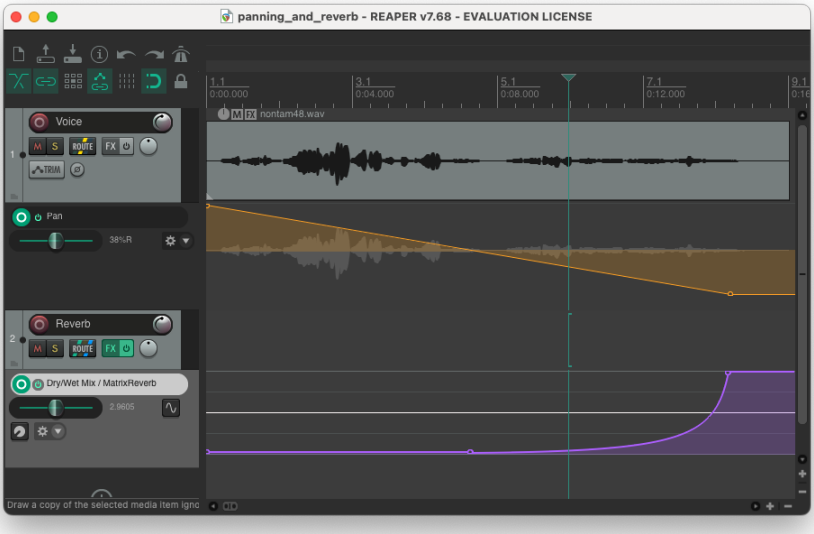
The assessment of how the two stereo channels interact is of prime importance when producing music, and quality issues that occur when the stereophonic recordings are mixed down into a monophonic signal for reproduction have been documented well before the advent of digital recording [32]. In more modern production, DAW plugins are generally used to visualize and assess the stereo qualities of the final mix [13, 33]. Figure 3.8b shows one example - the ToneBoosters Goniometer plugin [34]. It is designed to replicate an analogue goniometer, which, taking two voltage signals as input, displays them on a 2D axis such that a music producer might be able to discern the correlation between the signals.

In a large DAW session containing many tracks, the producer will make artistic choices about where in the stereo field to pan each track. While there are some typical practices in the industry, it must be emphasized that panning and creation of stereo field more generally is an artistic process [27], and the plugins that display stereo information usually serve as visual aides to notify the producer if their mix might exhibit a deterioration of the stereo field when played back in non-ideal reproduction settings.

3.7 Digital audio signals

When a sound wave is converted into an electrical signal by a microphone, it can be considered to be operating in continuous time. To convert the analogue signal into a digital one, it is sampled at regular intervals. The sampling frequency (f_s) is related to the sampling interval (Δ) by the relationship $\Delta = \frac{1}{f_s}$.

Shannon's sampling theorem states that a signal can be perfectly recreated when sampled at twice the highest frequency contained in the signal. Con-



(a) Recording project in the Reaper DAW [35]



(b) ToneBoosters Goniometer plugin [34]

Figure 3.8. Screenshots of example music production tools

versely, the highest frequency the highest frequency that can be represented by sampling a signal at a given sampling frequency is half of that frequency. This limit is called the 'Nyquist frequency' and is calculated by $f_N = \frac{f_s}{2}$ [36].

This is illustrated by Figure 3.9, which shows the sampling of three waveforms. The lower plot shows a signal at the Nyquist frequency (f_N), which can be perfectly reconstructed from the digital samples. The upper two waveforms have frequencies equidistant but in opposite directions from the Nyquist frequency ($f_N \pm f_\delta$). The digital samples shown in circles could represent either of the two waveforms. In practice, when the analogue waveform is reconstructed, the lower frequency will always take precedence.

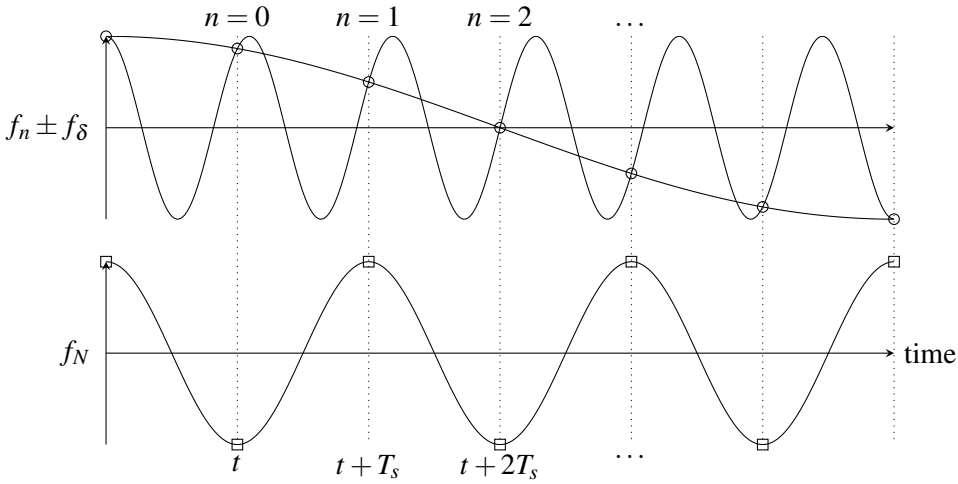


Figure 3.9. Aliasing in sampled digital audio signals. Dotted vertical lines represent digital sampling points with a separation of T_s, and monotonically increasing index n. The upper plot shows two cosine waves with frequencies equidistant from the Nyquist frequency (f_N) in opposing directions. The lower plot is a cosine wave at exactly f_N. Sampled values are shown as marks on the plots. Note that the two upper signals can both be represented by same digital samples.

3.8 Time-Frequency representation of audio

In order to perform many types of audio analysis, the audio signal can be converted from the time domain into the time-frequency (TF) domain. In perceptual acoustics research there are several common forms of TF representation, each with its own nuances. This section provides those details.

3.8.1 STFT spectrogram

The STFT spectrogram is one of the most popular TF represention across all audio processing work due to its computational efficiency. The fundamental principle is that a time-domain signal is split into short (1 to 100ms) overlapping chunks, and the Fast Fourier Transform (FFT) is performed on each chunk, thereby converting that chunk into the frequency domain. By arranging subsequent frames along a time axis it is possible to represent the original signal as a 2D image known as a spectrogram. When the time-domain signal is entirely real, (which is the case for monophonic audio signals), the result of the FFT will be a complex signal which is symmetrical around the Nyquist frequency and the upper half is therefore dropped. Furthermore, the complex spectrum is usually represented in magnitude/phase form to better match the human auditory response.

A fundamental property of the FFT is that it assumes that the signal repeats outside the time frame that is being analysed. Since this is generally not the case for audio signals, it is standard practice to apply a time-domain windowing function to each frame so that large transients do not occur at the boundaries. Serveral standard windowing functions exist, each with their own nuances, but for most audio applications, the 'Hann' window can be used [37].

The nature of the spectrogram is affected by two parameters: the frame size and hop size. 'Frame size' refers to the number of samples in each audio frame. To obtain maximal computational efficiency this must be a power of 2. 'Hop size' refers to the number of samples between the start of subsequent frames and is sometimes expressed as a percentage of the frame size. The frame size dictates the resolution of the spectrogram's frequency axis since the frequency bins will be spaced evenly between 0Hz and the Nyquist frequency. Importantly, a longer frame size will result in lower time accuracy within each frame; sharp transients in the time domain will appear as wideband energy in the frequency domain, but information about exactly where in the frame the transient occurred is lost. To counteract this inaccuracy, a smaller hop size can be used so that transients might be detected as sharp changes between STFT frames.

3.8.2 Mel spectrogram

One major drawback of the STFT spectrogram is that the frequency bins are spaced linearly from 0Hz to the Nyquist frequency, whereas the human perception of pitch works on a logarithmic scale. The STFT therefore contains redundant information in higher frequencies. One solution is to divide the entire spectrum into frequency bands that are spaced logarithmically within the entire spectrum and calculate weighted averages of the frequency bins within each band. The standard method is to use a 'Mel filter bank', an example of which is shown in fig. 3.10. When the Mel filter bank is applied to an STFT

spectrogram, the result is called a ‘Mel spectrogram’. An example is shown in fig. 3.11. Note that information in the higher frequencies has been condensed more-so that the lower frequencies.

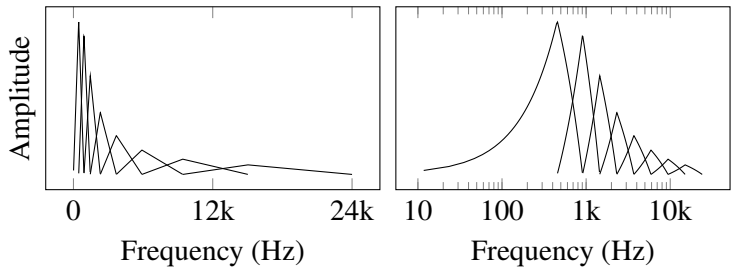


Figure 3.10. Example Mel filter bank containing 8 filters. Note that the bandwidth of each filter is roughly equal in logarithmic space. Since each filter is designed to give a weighted average of linearly spaced FFT bins, the amplitudes are scaled so that the sum of the weights in each band is equal.

3.8.3 Cochleagram

As an alternative to the two spectrogram representations, a cochleagram aims to make a TF representation that models the acoustic response of the human cochlea more closely. To that end, all processing is performed in the time domain and is sometimes referred to as a ‘peripheral ear model’. Several examples appear in the literature [38–40]. In the classical method [38] the original signal is first processed by a bank of time-domain gammatone filters with their centre frequencies placed according to a perceptual scale and each sub-band is subsequently processed by a half-wave rectifier, square root, and low-pass filter to mimic the excitation patterns of the cilia and basilar membrane. Each time-domain signal is then divided into overlapping time segments. The amplitude of each TF segment can then be calculated as the mean of all values its values, forming one pixel in the cochleagram. Figure 3.11 shows a comparison of all spectrogram types.

3.9 Machine learning

The large scale of the vast music repositories being addressed by the research questions naturally implies that some sort of machine learning is necessary.

Broadly speaking, machine learning techniques can be classified as either supervised or unsupervised. When training a supervised ML model, the training dataset needs to include ground-truth labels such that the network can objectively discern via a loss function whether the predictions it is making are

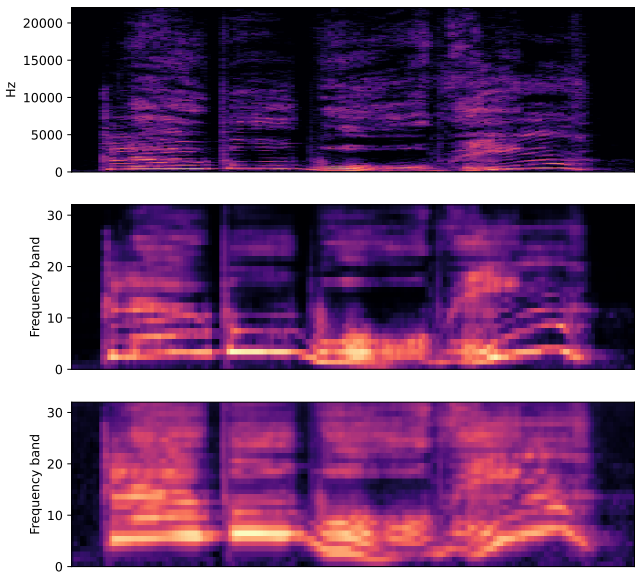


Figure 3.11. An example audio signal displayed as STFT spectrogram (above), Mel spectrogram (mid), and cochleagram (below).

correct. Unsupervised ML on the other hand does not require any labels. Instead, it seeks to form a representation of the underlying dataset via techniques such as clustering or dimensionality reduction, thereby learning the structures and patterns of data while retaining as much key information as possible.

Striking a middle ground between the two is 'self-supervised' learning, which addresses the issue of creating objective ground-truths from unlabelled data. Instead of relying on external labels, self-supervised learning generates pseudo-labels from the dataset itself, and then uses those as the input to a 'pretext task', which is tackled in much the same way as a supervised learning task.

As an example we can consider the current largest musical dataset: FMA [11]. The dataset contains the stereo audio of approximately 100 000 tracks, along with metadata such as genre tags. An unsupervised task might involve predicting the genre using the audio signal as input. A self-supervised task might involve masking off a short period of audio and filling it back in using the surrounding audio as context. An unsupervised task might involve learning a high-order representation of the raw spatial cues present in the stereo signals.

In relation to the overarching RQ in section 2.1, we assume that a music producer's DAW session is the only source of objective ground-truth data regarding stereo field. Furthermore, we assume that DAW sessions are not publicly available, as was found in [41, 42]; The only public-facing output of a recording session is the mixdown. It follows then that either unsupervised or self-supervised learning is necessary to draw any conclusions.

4. Related works

4.1 Spatial cues for stereo image

While the standard binaural cues mentioned in section 3.2 aim to anaylze binaural sound as perceived by humans, there are several examples of spatial cues taylored specifically to the analysis of recorded music. For example, the 'Panning Index (PI)' and 'ambience index' specified by Avendano and Jean-Marc [43], which are loosely analogous to ILD and IACC in the acoustics domain. To estimate the PI, a signal model is assumed whereby a musical source is panned within the mix using the 'energy preserving pan law', in which amplitude differences are applied to each channel such that the sum of the squared amplitudes equals 1. While this panning technique does not affect the phase of either channel, existing phase differences between channels are measured using the 'ambience index', which is defined such that it is low when the stereo channels are correlated (indicating a direct sound source), and high when channels are uncorrelated (indicating an ambient or reverberant sound source).

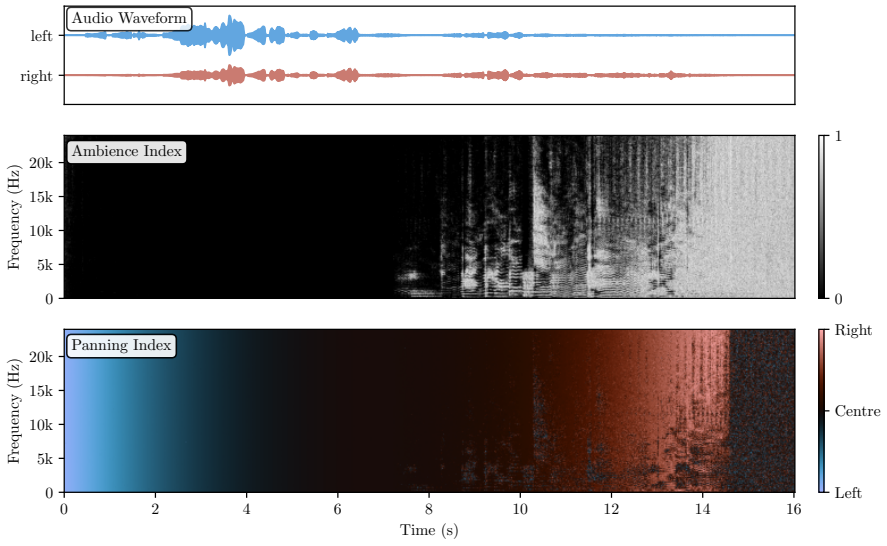


Figure 4.1. Visualization of an example audio waveform (top), its ambience index (middle), and panning index (bottom).

As the panning and ambience indices are key to many subsequent works, fig. 4.1 demonstrates them on an example music recording. The recording

itself is a solo singer recorded in an anechoic chamber, amplitude panned from left to right over the course of the musical phrase. At the midway point, an artificial reverb effect (i.e. a source of ambience) is added and slowly faded in. One can observe firstly that the ambience is low while there is no reverb, then increases steadily in frequency bands where audio energy is present, until finally all frequency bands have high ambience. Secondly, the panning index starts hard left and moves steadily right, becoming more and more unclear as the reverb is faded in.

Importantly, both the panning and ambience index are calculated in the TF domain and result in one value for each TF bin, and hence enable use cases that require resynthesis or editing of audio after analysis. One such example is the oft-cited Azimuth Discrimination and Resynthesis (ADRess) algorithm [44] (and its subsequent downstream projects [45–48]) (which attempt to separate sound sources by locating them in the horizontal plane). For the purpose of description, it is arguable that a summary of such spatial cues is more useful.

An extension and re-contextualization of the same principles has been discussed by Presti [49, 50], in which a TF region is converted into a Bivariate Mixture Space (BMS). Using this technique one can analyse the panning position of individual frequency components, plus the clarity (or trustworthyness) of that measurement. It was shown using this technique, one can e.g. differentiate direct from ambient sound, allowing for several audio manipulations and visualizations.

4.2 Stereo image description

FiXme: Would some examples of each existing SIP be useful?

Using the same signal model proposed by Avendano and Jean-Marc, the spatial description of recorded music specifically has been attempted several times. Firstly by Tzanetakis et al. in [6, 12], who use the PI to define a set of Stereo Panning Features (SPF) used for genre classification and MIR. SPF firstly splits the mix into perpetually relevant frequency bands, secondly describes the 'amount of panning' present in each band, and optionally estimates the number of sources present in the mix.

For a more detailed inspection of precisely where in the horizontal plane musical energy is located, one can instead utilize a Panning Histogram (PH) using the techniques described by Sarroff and Bello [8] or Gomez et al. [14].

While the aforementioned studies focus on descriptions of amplitude panning in stereo mixes, recording engineers can make use of other techniques to add spatial feeling to their mix. These techniques (described by Tardieu et al. as 'Phase Stereo') can be quantified using their definition of the Spectral Stereo Phase Spread (SSPS) [9]. In the same manuscript another method of quantifying amplitude panning is defined, which is based on the ILD of a

cochleagram, and aims to contain more precision in the frequency axis than the azimuthal axis.

An alternative signal model is proposed by Härmä [17, 18, 51], whereby a stereo mix is composed of combinations of centre, panned, and uncorrelated sources. Their subsequent classification of stereo field differentiates between different combinations of those source types.

Ziemer et al. argue the case in [13, 15, 52] that analyses based on a recording studio goniometer can aid in describing the stereo properties of music, in particular in differentiating between the artistic intentions of different recording sessions. The underlying logic is that since producers consult goniometers in their creative process, their readings must have inherent descriptive value. To perform the analyses, a virtual goniometer readout consisting of a lissajous figure and channel correlation measurement is constructed for non-overlapping time frames of an audio recording. The lissajous figure is quantized and reduced to a single number by counting the unique X/Y positions. Channel correlation is defined as the Pearson correlation coefficient [53] between the left and right channel signals.

All the above use classical DSP to create their spatial descriptors, but more recently there have been efforts to use DL to disentangle spatial information from audio content. One such example was demonstrated by Yang et al. [54]. While the original aim was to upmix stereo music to multichannel, a necessary component of the resulting neural network was a latent space that represents the spatial information of the multichannel signal. DL has also been used outside the music domain to e.g. learn spatial relationships between individual microphones in an array [55]. While such tasks do not directly address the research questions for this thesis, they demonstrate that learning spatial features on large amounts of unlabelled audio data is possible given appropriate methods and pretext task.

Neural generation of spatial audio continues to be a hot topic up to the present day, with many attempts to generate high-quality binaural, ambisonic or multichannel audio from mono sources [56–62]. In doing so, the networks must infer spatial information from a source other than audio (typically an accompanying video or text prompt), and in the process will learn the spatial qualities of their audio training data.

4.3 Prediction of perceptual scalars

A perceptual scalar is a measurement that would usually be made via a perceptual listening test, e.g. ASW or LEV. Prediction of them via objective means is a particularly useful function as it allows researchers to sidestep the tests, which can be slow and costly [63].

Acting purely on binaural signals the theses of Supper and Dewhirst ([7, 64]) details the construction of ‘binaural analysers’ that estimate the Direction

of Arrival (DOA) of incoming sound. Estimates of ASW is are derived downstream of DOA analysis, and involve taking standard measurements on DOA histograms. The analysis of stereo width more generally is a topic of interest and several mathematical formulas exists for its calculation [65–67]. It is not restricted to the binaural domain, also being calculated for generic stereo signals [8, 68, 69]. LEV on the other hand has been closely linked to the IACC of a stereo signal, and the prediction of it has been studied extensively in [70–75].

FiXme: Anything else about perceptual scalars? Maybe depth?

4.4 Spatial quality prediction

Following from the analyses described in sections 4.1 to 4.3, the results are often used to assess the spatial degradation of a signal after some sort of distortion has been applied. A typical method is to calculate the same spatial descriptor on both a reference and degraded signal, then calculate the difference in those descriptors. Examples are [76–81], which all attempt to predict the perceived spatial degradation in a recording that has been distorted by lossy compression.

FiXme: Very short. Is there anything more interesting to say?

4.5 Analysis of large musical datasets

Many of the studies cited in section 4.2 use a large dataset of recorded music to demonstrate the effectiveness of their SIPs. For example, [6, 9, 12] tackle genre recognition, [52] predicts artist/DJ, and [9] determines the decade of the recordings. Similar studies such as [82, 83] show that the geographic origin of a recording can be determined from audio features, although they include any spatial features.

In Panteli et al.'s work on the classification of singing style, the classes – instead of being determined by an a-priori known label – are learned by first extracting low-level audio cues, then finding common features within those cues. The features are then used as the input to unsupervised clustering algorithms to determine the relationships of the recordings to each other. This work was continued in [85] where outliers in the global dataset where found using similar techniques and links between different musical cultures were drawn. On a cautionary note, [86] recently showed that the choice of audio frontend (i.e. the choice of TF representation) implicitly encodes bias towards/against musical recordings that originate from different cultures.

What all these studies have in common is an interest in the dataset itself regarding both the relationships between individual recordings, plus the rela-

tionship of the entire corpus to the wider world. Much less emphasis is given to low-level audio features.

5. Research Methodology

Figure 5.1 shows a diagram of the research methodology for each paper included in this thesis and shows how each research track fed into subsequent tracks.

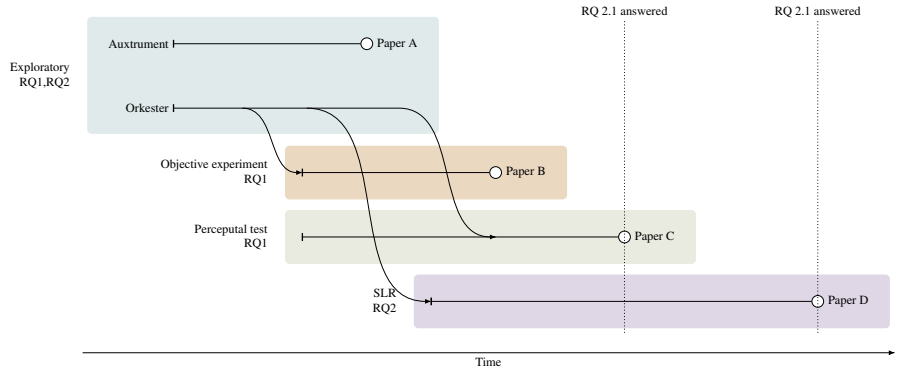


Figure 5.1. Diagram of research methodologies and their relation to published papers and research questions.

The research was initiated with a period of exploration and refinement of the research questions. Two work streams (named Auxtrument and Orkester) were established early on as frameworks to explore the importance of spatial information in musical audio, firstly from a musical performance perspective, and secondly from a signal processing perspective.

The Auxtrument initiative consisted of a joint project with many other researchers to make a musical performance that explored the use of space and merged realities. A custom instrument (the Auxtrument) was designed and built, the purpose of which was to generate a networked artificial reverberation chamber, embodying local sound sources with new spatial qualities [87, 88]. This resulted in the retrospective paper I [1].

Running concurrently to Auxtrument, the 'Orkester' initiative was an exploration of the more technical side of stereo field description. It involved a limited literature search to find methods of describing stereo field, followed by a period of software development to implement the discovered methods. Once the implementation was reliable, a period of musical exploration commenced, testing the various algorithms on recordings that were known to contain interesting spatial features. This had several outcomes. The most direct consequence was the inception of an objective experiment to determine how

the various stereo field description techniques reacted to different musical recordings, which culminated in paper II [2].

Concurrently to the objective experiment, a perceptual listening test was planned and executed to determine how listeners react to the panning position of speech overlayed on top of music. The software developed for Orkester was subsequently used to analyse the audio stimuli from the perceptual test in order to draw conclusions in paper III, which described the results and analysis of the perceptual test [3].

Lastly, through the developement of Orkester it became apparent that the original literature search was far from comprehensive. This spurred paper D, which was a systematic literature review designed to find all existing techniques that could be used to create a SIP [4].

The combination of the exploratory work, objective experiment, and perceptual test provided a satisfactory answer for RQ_1. The initial literature search plus the SLR provided an answer to RQ_2.

6. Paper summaries

Paper I

'Development of a Web-Based Real-Time Distributed Reverberation Chamber for Live Concert Applications'

Authors

Austin Franklin, Richard Mitic, Daniel Hedin, Rikard Lindell and Henrik Frisk

Status

Published in *Journal of The Audio Engineering Society*, [1].

Research methods

This paper presents a retrospective of a project to build a distributed audio performance over mobile networks. It follows the practice-based research methodology as defined by Candy [89].

Results and conclusions

The paper documents a real use case where stereo field and reproduction thereof is of prime importance. One conclusion was that network reliability was a limiting factor of the audio performance, sometimes leading to a degraded spatial impression. Research question 2.1 is addressed – Finding an adequate balance between audio spatial quality and network utilization would require tools to measure the stereo image.

My contribution

Member of the team that conceptualized the Auxtrument and made the initial survey and prototyping of existing audio transmission technologies. Responsible for post-prototyping of alternative approaches and custom audio software.

Paper II

'Towards an Objective Comparison of Panning Feature Algorithms for Unsupervised Learning'

Authors

Richard Mitic and Andreas Rossholm

Status

Published in *Proceedings of the 28th International Conference on Digital Audio Effects (DAFx25)*, September 2025

Research methods

Several DSP techniques for determining audio spatial features were performed on a small dataset that had predictable musical characteristics. The different techniques were compared using a new objective metric.

Results and conclusions

The paper revealed that although there are several DSP techniques specifically designed for the extraction of recording and panning features, they are all limited in their own ways. One particular technique (SPF from [6]) was particularly promising as a candidate for use later in the doctoral phase of studies, and the others could be used with some modifications. Overall it highlighted the importance of choosing both a panning feature DSP technique and a distance metric that are appropriate to the overarching task, thereby addressing research question 2.1.

My contribution

Main author, responsible for all data collection, experimental design, and software implementation. Presented the paper at the 28th International Conference on Digital Audio Effects in Ancona, Italy.

Paper III

'Perceptual Evaluation of Binaurally Panned Speech and Music Mixtures'

Authors

Richard Mitic, Majed Elwardy, Veronica Sundstedt, Andreas Rossholm and Yan Hu

Status

In review at *Acta Acustica*, March 2026.

Research methods

A perceptual test was used to determine listeners' reactions to carefully constructed audio stimuli. Standard statistical analysis was performed on the raw test results, and one particular DSP technique assessed in paper II (PH [14]) was used in an attempt to explain the perceptual results.

Remaining Work

Await peer review and address any comments that come up.

Results and conclusions

The paper concluded that when listeners are presented with voice overlayed on top of music, the best panning position for the voice was not always in the default centre position. The reasons for this were unclear due to a shortage of data, but the paper did demonstrate a real-world use of SIP extraction (using a modified algorithm trialled in paper II). Most relevant to the research question 2.1, it revealed that an objective measurement of panning can show detail that is not easily discernable by listening, such as very slight panning away from the centre.

My contribution

Joint main author alongside Majed Elwardy. Contributed to experiment design, data analysis, software implementation, and writing.

Paper IV

'A Systematic Literature Review of Methods to Describe the Stereo Image of Recorded Music'

Authors

Richard Mitic and Andreas Rossholm

Status

Accepted to *EURASIP Journal on Audio, Speech, and Music Processing*, May 2026

Research methods

Following on from the small selection of DSP techniques assessed in paper II, a SLR following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) standard was performed to gather all currently available methods for determining spatial features of stereophonic musical audio.

Results and conclusions

Addressing research question 2.1, the review revealed several more techniques that can be used to make spatial descriptions of stereo music. Despite the high number of studies that were assessed, there were relatively few 'very relevant' ones. The SLR also revealed that while ML/DL have been used on spatial features in the past, both *unsupervised* ML and very large music corpora remain fairly unexplored. It also details a framework for how to create a SIP algorithm that is useful for any given dataset and use case.

My contribution

Main author, responsible for all data collection and processing, and writing.

7. Discussion and future work

The work documented in this licentiate thesis provides the theoretical and practical basis for analysis of stereo image in music recordings. The focus was on amassing the signal processing techniques that enable the analysis of stereo field, and testing their applicability to various types of recordings that were known a-priori. The next stage of research will involve applying the knowledge from this thesis on much larger musical corpora. The signal processing techniques will be honed and run on vast musical datasets, secure in the knowledge that they will pick out the appropriate information to the experiment at hand.

In particular, it is theorized that examining the SIPs of extremely large musical corpora will reveal patterns in production and recording techniques that were otherwise unobservable. It will also reveal the scale and variation of stereo techniques that musicians are able to employ in their work. Given the basic knowledge gained from the work in this thesis, it might then be possible to reason about how musicians and producers sculpt their stereo fields, relating it both to the musical content and the wider context in which it was created.

During the licentiate, several points arose that provoked further investigation. Firstly, the available public datasets of recorded music are generally encoded using lossy compression. While lossy compression algorithms undergo perceptual tests in their development, it is unclear how they affect spatial cues at the scale of said datasets (which encompass much musical variation). The spatial effects of lossy compression therefore warrant further investigation. Secondly, the public datasets are generally formed from web services where musicians have uploaded their own recordings for free. In contrast, professional recordings (which are much more numerous) are not included in any public datasets. Since professional recordings make up the vast majority of recorded music, it needs to assessed whether public datasets are representative of professional recordings.

In short, the doctoral research will tackle a larger problem of assessing human musical creativity though the recorded output. The techniques surfaced by the licentiate will provide the foundation, and will be applied to real-world data.

Glossary

ADRess Azimuth Discrimination and Resynthesis 31
API Application Programming Interface 14
ASW Apparent Source Width 22, 32, 33

BMS Bivariate Mixture Space 31

DAW Digital Audio Workstation 23, 24, 29
DJ Disc Jockey 33
DL Deep Learning 11, 32, 40
DOA Direction of Arrival 32, 33
DSP Digital Signal Processing 17, 32, 38–40

FFT Fast Fourier Transform 26, 27
FMA Free Music Archive 11, 29

HRIR Head-Related Impulse Response 20, 21

IACC Interaural Cross Correlation 17, 30, 33
ILD Interaural Level Difference 17–19, 30, 31
IPD Interaural Phase Difference 18, 19
ITD Interaural Time Difference 17, 18

LEV Listener Envelopment 23, 32, 33

MIR Music Information Retrieval 13, 31
ML Machine Learning 11, 27, 29, 40

PH Panning Histogram 31, 39
PI Panning Index 30, 31
PRISMA Preferred Reporting Items for Systematic reviews and Meta-Analyses 40

RMS Root Mean Squared 19
RQ research question 13, 29

SIP Stereo Image Profile 14, 33, 36, 39–41
SLR Systematic Literature Review 14, 15, 35, 36, 40
SPF Stereo Panning Features 31, 38
SPL Sound Pressure Level 19
SSPS Spectral Stereo Phase Spread 31
STFT Short-time Fourier Transform ix, 26, 28

TF time-frequency 25–27, 31, 33

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